CAMELIA AL-NAJJAR

Department of Electrical, Computer, and Systems Engineering, Case Western Reserve University
10900 Euclid Avenue, Cleveland, OH 44106
caa20@case.edu | calnajjar.github.io
ORCID: 0009-0007-3348-9809 | Citizenship: United States

EDUCATION

Ph.D., Systems and Control Engineering, Case Western Reserve University Expected May 2027
Dissertation: "Choice Under Ambiguity and Other-Regarding Preferences in Games: Dispersion-Based Models with Out-of-Sample Evidence"
Committee: Behnam Malakooti (Chair), Vira Chankong, Roman Sheremeta, Daniel Solow

M.S., Computer Engineering, Texas A&M University, Fulbright Scholar 2005

B.S., Electrical Engineering, University of Jordan, Valedictorian 2002

FIELDS

Primary: Behavioral and Experimental Economics; Behavioral Game Theory
Secondary: Decision Theory; Mathematical and Quantitative Methods

JOB MARKET PAPER

"Game Geometric Dispersion Model with Comparative Out-of-Sample Empirical Study in Other-Regarding Preferences" (with Behnam Malakooti). Second-round revision, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.

The Nash equilibrium assumes that players care only about their own payoffs, yet experiments show that opponents' payoffs also shape choices. Existing other-regarding models are either linear in payoff differences or impose a single nonlinear form whether a player is ahead or behind. We propose the Game-Geometric Dispersion (G-GD) model, which applies separate nonlinear adjustments to the advantaged and disadvantaged branches based on the geometric dispersion of all players' payoffs. With two parameters, G-GD represents self-interest and four types of other-regarding preferences, and it adds negligible cost to computing the equilibrium. Using 80 observed choice distributions in three datasets from four published experiments, we compare G-GD with the Nash equilibrium and three leading other-regarding models in repeated out-of-sample tests. G-GD attains the highest mean out-of-sample log-likelihood in all three datasets, and its advantage is significant at the 5% level. Estimated preference types differ by game category, suggesting that other-regarding behavior depends on strategic context.

Online supplement with replication code, data, and results: SSRN, doi:10.2139/ssrn.6372538

PUBLICATIONS

"Systems Analysis of Decision-Making Under Ambiguity with Comparative Out-of-Sample Experimental Study of Geometric Dispersion Theory and Cumulative Prospect Theory" (with Behnam Malakooti). *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 55(8): 5192–5206, 2025.

Introduces Ambiguity GDT (A-GDT), which out-predicts CPT, SEU, α-maxmin EU, and vector EU on held-out choices in an incentivized experiment (N = 310, 150 choices per subject).

"Basic Geometric Dispersion Theory of Decision Making Under Risk: Asymmetric Risk Relativity, New Predictions of Empirical Behaviors, and Risk Triad" (with Behnam Malakooti and Mohamed Komaki). *Decision Analysis*, 18(1): 41–77, 2021.

WORKING PAPERS

"Geometric Dispersion Explanation of Prisoner's Dilemma with Behavioral Out-of-Sample Comparisons to Game Models" (with Behnam Malakooti), 2026.

Applies G-GD to the one-shot Prisoner's Dilemma. G-GD predicts cooperation rates out of sample better than Nash equilibrium, quantal response equilibrium, and the Fehr–Schmidt, Charness–Rabin, and Bolton–Ockenfels models.

RESEARCH IN PROGRESS

Treatment choice under ambiguity (planned): field test of A-GDT on treatment-plan acceptance.

Heterogeneity across domains (planned): within-subject risk, ambiguity, and social preferences.

Model selection for behavioral theories (planned): out-of-sample protocol on public datasets.

DATA AND REPLICATION MATERIALS

"A New Dataset on Eliciting Behavioral Data on Ambiguity in Monetary Decisions" (with Behnam Malakooti). IEEE DataPort, 2023. doi:10.21227/a8qn-d389. Incentivized MTurk/Qualtrics experiment with Stata replication code.

"Online Supplement for: Game Geometric Dispersion Model with Comparative Out-of-Sample Empirical Study in Other-Regarding Preferences" (with Behnam Malakooti). SSRN, 2026. doi:10.2139/ssrn.6372538. Estimation code, data, and out-of-sample results for the job market paper.

OTHER PUBLICATIONS (OPTIMIZATION AND ENGINEERING)

"Multi-Objective Energy Aware Multiprocessor Scheduling Using Bat Intelligence" (with Behnam Malakooti, Shaya Sheikh, and Hyun Kim). *Journal of Intelligent Manufacturing*, 24(4): 805–819, 2013.

"Hybrid-LP: Finding Advanced Starting Points for Simplex, and Pivoting LP Methods" (with Behnam Malakooti). *Computers & Operations Research*, 38(2): 427–434, 2011.

"Multi-Objective OLSR for Proactive Routing in MANET with Delay, Energy, and Link Lifetime Predictions" (with Zhihao Guo, Shahdi Malakooti, Shaya Sheikh, and Behnam Malakooti). *Applied Mathematical Modelling*, 35(3): 1413–1426, 2011.

"Energy Aware Proactive Optimized Link State Routing in Mobile Ad-Hoc Networks" (with Zhihao Guo, Shahdi Malakooti, Shaya Sheikh, Matthew Lehman, and Behnam Malakooti). *Applied Mathematical Modelling*, 35(10): 4715–4729, 2011.

"Mobile Ad Hoc Network Broadcasting: A Multi-Criteria Approach" (with Brian J. Barritt, Shaya Sheikh, and Behnam Malakooti). *International Journal of Communication Systems*, 24(4): 438–460, 2011.

"The Complex Interior-Boundary Method for Linear and Nonlinear Programming with Linear Constraints" (with Behnam Malakooti). *Applied Mathematics and Computation*, 216(7): 1903–1917, 2010.

"Mobile Ad Hoc Network Proactive Routing with Delay Prediction Using Neural Network" (with Zhihao Guo, Shaya Sheikh, Hyun Kim, and Behnam Malakooti). *Wireless Networks*, 16(6): 1601–1620, 2010.

"A Class of Models for Microgrid Optimization" (with Shaya Sheikh, Mohamed Komaki, Abdulaziz Altowijri, and Behnam Malakooti). In R. Ramanathan, M. Mathirajan, and A. R. Ravindran (eds.), *Big Data Analytics Using Multiple Criteria Decision-Making Models*, CRC Press, 2017, pp. 155–174.

"A Service Provider's Approach for Improving Performance of Aggregate Voice-over-IP Traffic" (with A. L. Narasimha Reddy). *International Workshop on Quality of Service (IWQoS)*, 2006.

TEACHING EXPERIENCE

Teaching Assistant, EECS 346 Engineering Optimization, CWRU — Fall 2014
Delivered two guest lectures; graded homework and exams.

Design Instructor, EECS 398/399 Senior Project, CWRU — 2012–2014
Led progress reviews and class discussions; evaluated reports and interim presentations.

Faculty Lab Instructor, Department of Electrical Engineering, University of Jordan — 2002–2003
Primary instructor for the electrical circuits and logic circuits laboratories; wrote weekly quizzes and final exam questions.

Certification: UNIV 401, Advanced Professional Development for University Teaching — 2020

Teaching fields: Decision Theory; Behavioral Economics; Game Theory; Mathematical Methods and Optimization; Quantitative Methods (Stata, R)

RESEARCH MENTORING

Undergraduate senior project, predictive modeling of Type I diabetes, CWRU — 2008
Co-supervised with Behnam Malakooti; resulting poster presented at ShowCASE 2009.

HONORS AND FELLOWSHIPS

Fulbright Scholarship (full funding, M.S. Computer Engineering) — 2003–2005
Phi Kappa Phi Honor Society — 2005
Ranked first, College of Engineering, University of Jordan — 2002

PROFESSIONAL SERVICE

Judge, CWRU Intersections Poster Symposium — 2024, 2025
Referee, *Journal of Risk and Uncertainty* — 2026
Memberships: INFORMS, IEEE

RESEARCH SKILLS

Estimation and programming: Stata (custom maximum likelihood), R, MATLAB, Python, LINGO, Mathematica, SQL (MySQL), C/C++
Experimental methods: Qualtrics, Amazon Mechanical Turk, incentive-compatible elicitation

OTHER PROFESSIONAL EXPERIENCE

Operations and Technology Manager, York Dental (family business), Cleveland, OH — 2006–present
Electrical Engineering Intern, Desktop BIOS Group, Dell Inc., Austin, TX — 2004

REFERENCES

Behnam Malakooti (Advisor)
Professor of Electrical, Computer, and Systems Engineering
Case Western Reserve University
behnam.malakooti@case.edu

Roman Sheremeta
Associate Professor of Economics, Weatherhead School of Management
Case Western Reserve University
roman.sheremeta@case.edu

Vira Chankong
Professor of Electrical, Computer, and Systems Engineering
Case Western Reserve University
vira.chankong@case.edu